

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from:</i> • Discharge of tank-washing water from tankers • Oil rig / drilling accidents / leakage during drilling • Pipeline leaks • Leakage from storage tanks / refinery spills • Waste lubricating oil from vehicles / industry washed into drains • Other ship accidents (fuel oil)	3
01.2	• Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia • Feathers lose buoyancy / waterproofing so heavily oiled birds drown • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young	3
01.3	<i>Any 2 from:</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen • Toxic components (benzene, xylene) dissolve and poison plankton near the surface • Oil blocks light so algae / phytoplankton cannot photosynthesise • Oil masks odours / narcotises so animals cannot find mates or food by smell • Shore organisms (molluscs, crustaceans) are covered and asphyxiated	2
01.4	• Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	

Question	Marking guidance	Marks
02.1	• LEO: high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover • But each area is only revisited every 15 days so they cannot follow rapidly changing weather • Geostationary: constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower	3
02.2	<i>Any 3 from:</i> • Wind velocity • Ocean currents / sea surface temperature • Wave height • Ice thickness • Vegetation cover • Sea level • Atmospheric temperature / composition	3
02.3	• Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome • If the prediction matches, the model can be trusted (with caution); it is continually modified as more data are collected and differences analysed	2
02.4	• Models need data from all regions / oceans / altitudes; gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions • Unrepresentative data (eg urban heat islands, short time series) would bias the model; remote sensing helps fill gaps	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
03.1	• Orientation: south-facing windows (northern hemisphere) maximise passive solar gains; rooms needing warmth on the sunny side; fewer / smaller windows on the north side where losses are greatest; solar screens / brise soleil prevent summer overheating • Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost • Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced	6
03.2	• Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy • Limecrete has about half the embodied energy of concrete; rammed earth, straw bales and lime mortar are lower still	2
03.3	• Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area • Heat transfers from the outgoing to the incoming air so fresh air enters already warmed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	

Question	Marking guidance	Marks
04.1	<i>Any 3 from:</i> • Disposal of items containing mercury: batteries, thermometers, fluorescent lights • Chemical plants producing chlorine using mercury electrodes • Combustion of coal • Gold mining (amalgamation)	3
04.2	<i>Any 3 from:</i> • Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) • Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters • Disposal / storage of wastes at high pH to reduce solubility • Encapsulation of mercury-containing hazardous waste in cement	3
04.3	• Most heavy metals are more soluble at low pH / less soluble at high pH • So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms	2
04.4	• The effects of two pollutants interact to produce a different (greater) effect than either alone • Cadmium and zinc; ozone and sulfur dioxide	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
05.1	• Landfill fell from 79% to 14% (large decrease) • Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest)	2
05.2	<i>Any 3 from:</i> • Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases • Toxic leachate may leak into groundwater / rivers from poorly managed sites • Uses large areas of land; loss of habitats / farmland; contamination prevents later use • Resource value of metals, glass, plastics is lost • Transport noise / congestion / litter / pests / odour	3
05.3	<i>Any 4 from:</i> • Polymer / clay liner to prevent escape of leachate • Collection and treatment of leachate • Collection of methane and use as a fuel • Impermeable cap (clay and polymer) when complete to stop water entering • Regular covering with soil (pests, odour); deodorising sprays • Perimeter fence to trap wind-blown litter • Separation and recording of waste types; reactive / flammable wastes not dumped together	4
05.4	• Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
06.1	• (60 x 45) / 9 • = 300	2

Question	Marking guidance	Marks
06.2	<i>Any 4 from:</i> • Marked individuals mix freely / randomly with the rest of the population • No births, deaths, immigration or emigration between samples • Marking does not affect survival or behaviour (eg make them more visible to predators) • Marks do not wear off / fall off • All individuals have the same chance of being caught (traps do not attract or repel marked animals)	4
06.3	• Fewer marked beetles would survive to be recaptured • So the proportion marked in the second sample would be too low and the population would be overestimated	2
06.4	• Enough time for marked beetles to mix with the population • Not so long that births / deaths / migration or loss of marks change the population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
07.1	• Potable: no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints • Spray irrigation: low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food • Condenser cooling: no gross solids - pipe blockage • Boiler water: no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes	4
07.2	• Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter • Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed	2
07.3	• Clay particles carry electrostatic charges that make them repel each other so they stay suspended • Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier	2
07.4	<i>Any 2 from:</i> • Chlorine • Ozone • Ultraviolet light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
08.1	• B - tiny pores between clay particles allow capillary rise; none in sand	1
08.2	• Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants • Gravitational: drains freely through large pores; available to plants only briefly as it passes	3
08.3	• Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Question	Marking guidance	Marks
09.1	• Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 • Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight	2
09.2	• 24-hour day / night cycle: people asleep overnight; morning rise as heating, lighting, cooking, industry and offices start • Evening peak: lighting, cooking, heating as people return home; short-term 'TV pickup' at programme breaks / mealtimes • Other causes: seasonal (winter heating / lighting, summer air conditioning); weather-related (cold snaps); weekday / weekend differences in industrial use	4
09.3	• They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly • Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables	2
09.4	<i>Any 2 from:</i> • Intermittent renewable resources: wind, solar, tidal • Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) • Breakdowns / maintenance of power stations	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts • Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating • HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital • Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact • Wave: long Atlantic fetch, but technology immature • Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat • Geothermal: limited to low-temperature schemes / heat pumps • Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
10.2	• Eight times (energy proportional to velocity cubed)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Nature of the problem: interconnected atmosphere-hydrosphere-biosphere systems; changes over different time and spatial scales; natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change; time delay between cause and effect (ocean heat capacity) • Limitations of historic data: sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • Proxy data and their reliability: dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • Modern monitoring: long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty; ecological uncertainty (species survival, distribution, fragmentation) • Computer models: how they are tested and refined; importance of accurate representative data; scenario-based predictions and ranges • Uncertainty over ENSO / Conveyor / jet stream responses • Evaluation and conclusion: confidence in the overall trend and human cause despite detailed uncertainty; role of peer review and international bodies (IPCC) Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25